

Strategic Planning of Agroforestry Supply Chains

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Abstract

Agroforestry systems that integrate fast-growing woody perennials into agricultural land form a structurally novel type of biological-industrial symbiosis: they simultaneously deliver multiple biomass fractions of differing quality to chemical, pulp-and-paper, and energy industries through a shared regional supply network, thereby substituting fossil feedstocks and storing biogenic carbon in long-lived products. Despite this environmental potential, the material flows through such networks are poorly understood because stand-level age dynamics, multi-fraction biomass composition, and product quality cascading across industrial sectors have never been jointly optimised. This paper presents the AFS-SCD model, a mixed-integer linear programming (MILP) formulation that endogenises agroforestry establishment timing and consecutive harvest scheduling through arc-based age-lag constraints, couples component-specific Gompertz yield functions for stem, branch, and residue fractions with a multi-echelon network design model, and implements a product quality cascade allocating biomass grades to chemical, pulp, and energy consumers within a unified framework. Applied to a real-world case study in the Saxony-Anhalt tri-state region of Germany — a cluster of chemical and paper industries seeking bio-based feedstock transition — the model identifies the profit-optimal spatial configuration, rotation lengths, and product allocation across a 40-year planning horizon. Sensitivity analyses across opportunity-cost, logistics-cost, and revenue scenarios reveal that product cascading is the dominant lever for AFS viability, that sites with opportunity costs above approximately 300 EUR per hectare per year are rarely activated, and that high-value stem allocation to chemical applications substantially outperforms energy-only offtake strategies. The findings demonstrate that integrated optimisation of biological production dynamics and industrial network design is essential for designing viable and cleaner agroforestry bioeconomy ecosystems.

Keywords: Industrial symbiosis, Agroforestry, Biomass cascading, Bioeconomy network,

30 **1. Introduction**

31 The industrial ecology of biobased economies depends on one central challenge: con-
32 necting primary biological production systems to multiple industrial sectors through
33 networks that efficiently allocate biomass fractions of varying quality to their highest-
34 value application — the principle of industrial symbiosis applied to renewable material
35 flows (Hildebrandt et al., 2019). Agroforestry systems (AFS) that deliberately integrate
36 fast-growing woody perennials into agricultural land represent a structurally novel real-
37 isation of this principle. By co-producing stem wood suitable for chemical and paper
38 industry, branch wood suitable for pulping, and residue fractions suitable for energy
39 generation, AFS create a multi-fraction feedstock that can simultaneously serve multiple
40 industrial consumer sectors through a shared regional supply network (Toensmeier, 2017).
41 This biological-industrial symbiosis substitutes fossil feedstocks with renewable bio-based
42 alternatives and stores biogenic carbon in long-lived material products — a cascade strat-
43 egy explicitly preferred over energetic use from a cleaner production perspective (Huber
44 et al., 2018).

45 Despite their ecological and industrial symbiosis potential, temperate AFS occupy
46 only a marginal share of agricultural land in practice. In Germany, approximately 1,200
47 hectares of silvoarable AFS were established in 2025, accounting for barely 0.02% of
48 total agricultural land (Deutscher Fachverband für Agroforstwirtschaft (DeFAF) e.V.,
49 2026). This gap between potential and practice reflects an unresolved material flow
50 planning problem: establishing AFS at regional scale requires coordinated decisions on
51 stand establishment timing, multi-cycle harvest scheduling, and biomass fraction alloca-
52 tion across spatially distributed industrial consumers — decisions that interact through
53 the age-dependent composition of harvested biomass. At short rotation ages, harvested
54 biomass is dominated by residue and branch fractions of limited merchantability; at
55 longer rotations, stem diameters surpass quality thresholds and the share of chemical-

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56 and paper-grade material rises markedly (Huber et al., 2018; Niemczyk and Thomas,
57 2021). The material flow structure of an AFS-based bioeconomy network therefore de-
58 pends endogenously on the rotation strategy, which cannot be set independently of the
59 product cascade and the industrial demand structure. Current biomass supply chain
60 models, reviewed in Section 2.4, do not capture this coupling: they either treat biomass
61 availability as an exogenous time series — decoupling it from age-class dynamics — or
62 optimise a single downstream value chain rather than a multi-sector product cascade
63 (Zahraee et al., 2020; De Meyer et al., 2016).

64 This paper closes that gap by developing the **AFS Supply Chain Design (AFS-**
65 **SCD)** model, a mixed-integer linear programming (MILP) formulation for the integrated
66 design and operation of an agroforestry biomass supply chain from an industrial ecology
67 perspective. The model makes three contributions to the industrial ecology of biobased
68 networks. First, it endogenises the material flow structure of an AFS network by coupling
69 age-class stand dynamics — formalised through binary arc-based age-lag variables that
70 enforce minimum and maximum rotation constraints at each site — directly to regional
71 logistics decisions, so that rotation length, harvest timing, and network configuration
72 are optimised jointly. Second, it embeds component-specific Gompertz yield functions
73 for stem, branch, and residue fractions, translating stand age into merchantable dry-
74 matter quantities per hectare for each biomass type, thereby making the multi-fraction
75 material flow endogenous to the optimisation. Third, it implements a product quality
76 cascade across three industrial sectors — chemical/biorefinery (P1), pulp and paper (P2),
77 and energy/biogas (P3) — within a unified multi-period network design model including
78 storage, processing, and inter-sector substitution. The model is applied to a real-world
79 industrial symbiosis context in the Saxony-Anhalt tri-state region of Germany, where
80 a dense cluster of chemical and paper industries is undergoing a bio-based feedstock
81 transition and seeks reliable regional woody biomass supply.

82 The remainder of the paper is structured as follows. Section 2 reviews the relevant
83 literature on agroforestry systems, biomass growth modelling, and biomass supply chain
84 design, concluding with the identification of the modelling gap addressed here. Section
85 3 presents the AFS-SCD formulation. Section 4 describes the Saxony-Anhalt case study,
86 presents the base-case optimisation results, and analyses the implications of alternative

87 cost and revenue scenarios for regional cleaner production strategies. Section ?? con-
88 cludes and discusses directions for integrating life cycle assessment into the framework.

89 2. Literature review

90 Agroforestry biomass systems, stand-level growth models, and biomass supply chain
91 optimisation have each been studied extensively, but they are rarely combined in an inte-
92 grated framework that links age-dependent stand dynamics to regional product-cascading
93 value chains (Zahraee et al., 2020; Sharma et al., 2013; Espinoza et al., 2017). This sec-
94 tion reviews (i) agroforestry systems with emphasis on system types, economics, and the
95 cascading use of biomass as the primary revenue driver; (ii) biomass growth models at
96 tree and stand level with explicit attention to the allometric–diameter–stand-biomass
97 pathway; and (iii) biomass supply chain design using mathematical programming. The
98 section concludes by identifying the modelling gap addressed in this paper.

99 2.1. Agroforestry systems

100 Agroforestry deliberately integrates woody perennials with crops and/or livestock on
101 the same management unit, with the aim of enhancing productivity, profitability, and
102 environmental performance (Nair, 1993; Toensmeier, 2017). In the temperate zone, three
103 distinct system types can be distinguished based on the spatial arrangement of trees and
104 the rotation length at which woody biomass is harvested.

105 **Short-rotation coppice (SRC)** is defined as a monoculture plantation of fast-
106 growing woody species covering the entire management unit, with planting densities
107 of 8,000–20,000 cuttings·ha⁻¹. After each harvest the root system regenerates without
108 replanting. Rotation cycles are typically 3–7~years and total stand lifetime is 20–25~years
109 (Trnka et al., 2008; Faasch and Patenaude, 2012). SRCs are no AFSs in the narrow sense,
110 but it is helpful to benchmark biomass production.

111 **Short-rotation agroforestry systems (SRAFS)** integrate SRC-type tree strips
112 (typically 1–4 rows of the same fast-growing species like poplar or willow) into an oth-
113 erwise arable field in an alley-cropping arrangement. Crop production continues in the
114 inter-row strips (Huber et al., 2018; Graves et al., 2007). Thus, tree strips typically cover
115 10–30% of the total parcel area. Rotation lengths and total stand lifetime in SRAFS are

comparable to SRCs. In SRAFS all trees are effectively edge trees, exposed to higher light and water availability than interior SRC trees. This boundary effect raises individual-tree biomass accumulation but also intensifies competition with adjacent crop rows (Huber et al., 2018).

Long-rotation agroforestry systems (LRAFS) use the same alley-cropping spatial structure as SRAFS but with rotation cycles of 12–15~years or longer (Graves et al., 2007; Nair, 1993). At this rotation length, individual trees develop substantial stem diameters and the proportion of high-value timber increases markedly, enabling premium material uses (veneer, sawlogs, engineered wood products) that are inaccessible in short-rotation systems. Therefore, often timber-quality tree species (such as walnut, cherry, oak, or high-grade poplar clones) are planted.

SRAFS and LRAFS are not clearly differentiated in practice. First, a clear cut-off age is hard to define and, more importantly, the intended use of AFS biomass is not always clear and can also change over time. Nonetheless, when establishing AFS, the structure of AFS and its intended use after harvest drives planting decisions and, subsequently, economic viability. Thus, in the present paper SRAFS and LRAFS are not considered as separate AFS systems, but the interplay of intended use, establishment, and harvesting decisions is analysed.

2.2. Economic rationale of landowners

The decision of a landowner to establish an AFS involves weighing substantial upfront costs and long-term opportunity costs against a diversified revenue stream. On the cost side, establishment requires site preparation, planting material, and planting labour, typically amounting to 1,500–3,500 €/ha for SRC or SRAFS (Faasch and Patenaude, 2012; Testa et al., 2014). Recurrent harvest and refitting costs (machine hire, chipping, transport to roadside) add approximately 200–400 €/ha per rotation depending on stand density and equipment availability (Testa et al., 2014; Spinelli et al., 2009). During the rotation, maintenance costs of AFS arise on a yearly basis. Next to expenses for establishment, maintenance and harvesting, opportunity costs arise as no revenues from crops are generated on tree strips. In regions with fertile soils and high cereal prices, yearly crop revenues may sum up to 400–600 €/ha (Faasch and Patenaude, 2012; Graves

et al., 2007). Thus, opportunity costs are often the dominant cost driver in SRAFS profitability analyses.

On the **benefit side**, the primary revenue of AFS is the sale of harvested woody biomass, whose magnitude depends critically on rotation length, species, and site productivity. For poplar-based SRAFS in Central Europe, typical yields range from 10 to 25 t of dry matter biomass per ha for typical rotation lengths (Huber et al., 2018; Niemczyk and Thomas, 2021). Additionally, positive crop-yield effects in the inter-row strips of SRAFS and LRAFS may result. Tree rows act as windbreaks, reducing soil erosion and evapotranspiration; they provide partial shading that moderates heat stress during drought years; and their root systems may improve water retention in the topsoil (Graves et al., 2007; Grünewald et al., 2007). Empirical measurements in Central European alley-cropping trials indicate that these microclimate effects can increase cereal yields in adjacent strips by 5–15% relative to open-field controls, partially or fully compensating the loss of arable area occupied by the tree strips (Graves et al., 2007). Beyond direct yield effects, AFS deliver broader ecosystem services (like soil organic matter accumulation, carbon sequestration, biodiversity enhancement) that may lead to agri-environment subsidies and carbon credits (Toensmeier, 2017; Faasch and Patenaude, 2012).

Nonetheless, economic attractiveness of AFS depends primarily on revenues of AFS biomass. Thereby, revenues are different for the types of biomass and in which value chain it is used. Often AFS biomasses are used energetically, i.e. for heat and power generation, or anaerobic digestion, which typically creates low prices per tonne of biomass.[REF] Alternatively, woody parts of the biomass (i.e., the stem and large branches) can also be used as pulp and paper grade material as well as feedstock for chemical industry or as timber achieving higher prices (Lamers et al., 2011) [EXTEND REFS]. Thus, fostering AFS establishment by maximizing economic returns from AFS biomass requires a strategic approach to product cascading that allocates different biomass fractions to the highest-value applications, while ensuring that the rotation length is aligned with the growth dynamics of the stand and the market demand for different biomass types (Lamers et al., 2011).

Thus, for making educated decisions on AFS establishment and harvesting, the ex-

pected biomass quantities at harvest for stem, branches, and residues need to be estimated. Thereby, the shares of the biomass types change with stand age. E.g.~the share of stem weight rises from roughly 55% at age 4 to over 75% at age 10 (Huber et al., 2018; Testa et al., 2014). Therefore, in Section 2.3 growth modelling approaches are reviewed that capture the age-dependent growth dynamics of different biomass types in AFS.

2.3. Biomass growth models

Reliable estimation of stand-level biomass quantities and their age-dependent composition is a necessary prerequisite for any strategic analysis of AFS-based supply chains. Individual-tree biomass is commonly approximated by species-specific allometric power laws relating shoot dry matter to stem base diameter, and these relationships have been calibrated for Central European poplar clones under SRC and SRAFS conditions (Huber et al., 2016, 2018). For supply chain planning, however, the relevant quantity is stand-level biomass per hectare over a multi-decade rotation horizon, which requires a temporal growth model that captures the characteristic sigmoidal accumulation trajectory of woody biomass. Therefore, a set of variables is used summarized in ??.

2.3.1. Stand-level growth model

Stand-level aboveground and belowground dry biomass per hectare, $M(t)$, accumulates in a right-skewed sigmoidal pattern with a relatively early inflection point, driven by rapid juvenile shoot growth followed by self-thinning and competitive deceleration as canopy closure intensifies (Niemczyk and Thomas, 2021; Testa et al., 2014). This trajectory is described by a Gompertz model:

$$M(t) = A(N) \cdot \exp(-\exp(-k \cdot (t - t_0))), \quad (1)$$

where t_0 is the age of maximum instantaneous biomass increment and k determines how quickly the stand approaches its site-specific asymptote $A(N)$. For Central European poplar under medium-rotation AFS management (10–20 yr), analysing empirical data from Huber et al. (2018), Niemczyk and Thomas (2021), and Jha (2018) yields parameter intervals of $k \in [0.17, 0.2]$ and $t_0 \in [9, 10]$ years, depending on local specifications of the AFS under study. Parameter t_0 indicates an upper bound on the optimal rotation age as growth slows down or diminishes entirely close to this age.

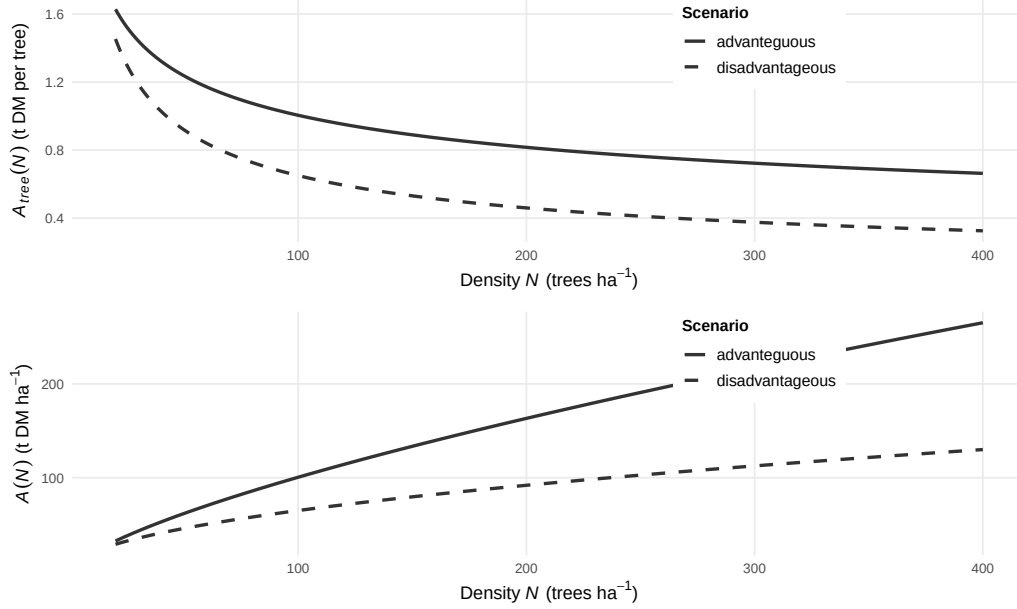


Figure 1: Tree-level and stand-level Gompertz asymptote functions $A_{tree}(N)$ and $A(N)$ for advantageous ($N = 4$, $\beta = -0.3$) and disadvantageous ($N = 6.5$, $\beta = -0.5$) site conditions.

205 The asymptote $A(N)$ depends largely on planting density and can be modelled by a
 206 competition-density power law (Kira et al., 1961):

$$A(N) = C_{\text{site}} \cdot N^{1-\beta}, \quad (2)$$

207 where C_{site} encodes site quality, N indicates tree density (e.g. trees per hectare), and β is
 208 the competition exponent. Equivalently, the per-tree asymptote $A_{\text{tree}}(N) = A(N)/N =$
 209 $C_{\text{site}} \cdot N^{-\beta}$ decreases with density as individual trees have access to less growing space.
 210 Parameters C_{site} and β are highly site- and species-specific. For Poplar clones under
 211 European conditions, $\beta \in [0.3, 0.5]$ and $C_{\text{site}} \in [4, 6.5]$ (Niemczyk and Thomas, 2021, Kira
 212 et al. (1961)). Figure 1 shows asymptote functions according to (2) for advantageous and
 213 disadvantageous site conditions.

214 When assessing potential downstream application of AFS biomass, biomass fraction
 215 are to be defined meeting the requirements of downstream consumers. E.g. for timber
 216 and biochemical applications, stem diameters of AFS trees need to exceed a minimum
 217 threshold in order to qualify as a suitable feedstock. In the following three biomass
 218 fractions are distinguished: stems, branch wood, and residue. The stem fraction consists

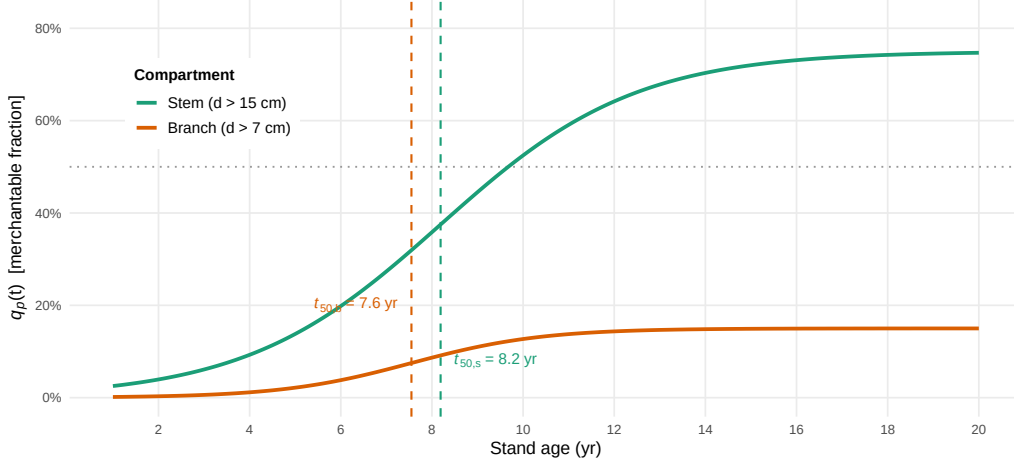


Figure 2: Logistic diameter-class merchantability functions $q_p(t)$ for stem ($d > 15$ cm, green) and branches ($d > 7$ cm, orange); the 50% threshold ages t_{50} are indicated by dashed vertical lines.

of stems with a diameter of at least 15 cm, while branch wood requires a diameter of at least 7 cm. The residue fraction comprises are smaller parts of the tree. To subdivide the total AFS biomass modelled by (1) into the three fractions depending on tree age, a logistic growth model is used:

$$q_p(t) = \frac{f_p}{1 + \exp(-r_p \cdot (t - t_{50,p}))}, \quad (3)$$

where $t_{50,p}$ is the age at which 50% of compartment p biomass surpasses the diameter threshold and f_p denotes the asymptotic share of component p for mature trees. Using empirical data from data from Civitarese et al. (2019), Jha (2018) and Niemczyk and Thomas (2021), $r_{stem} = 0.467, \text{yr}^{-1}$, $t_{50,stem} = 8.19$ years, and $f_{stem} = 0.75$ is estimated for the stem fraction (threshold $d > 15$ cm) and $r_{bran} = 0.698 \text{ yr}^{-1}$, $t_{50,bran} = 7.55, \text{yr}$, and $f_{bran} = 0.15$ for the branch fraction (diameter at least 7 cm at insertion). Figure 2 illustrates the development of the fraction shares over stand age.

Combining equations (1)–(3), the dry biomass yield of fraction p per hectare at harvest age t is:

$$\eta_p(t) = q_p(t) \cdot M(t), \quad (4)$$

with all non-merchantable fractions aggregated into the residue fraction. Fresh-weight yields at felling are obtained from dry matter yields via $\eta_p^{\text{fw}}(t) = \eta_p(t)/(1 - \omega)$, where

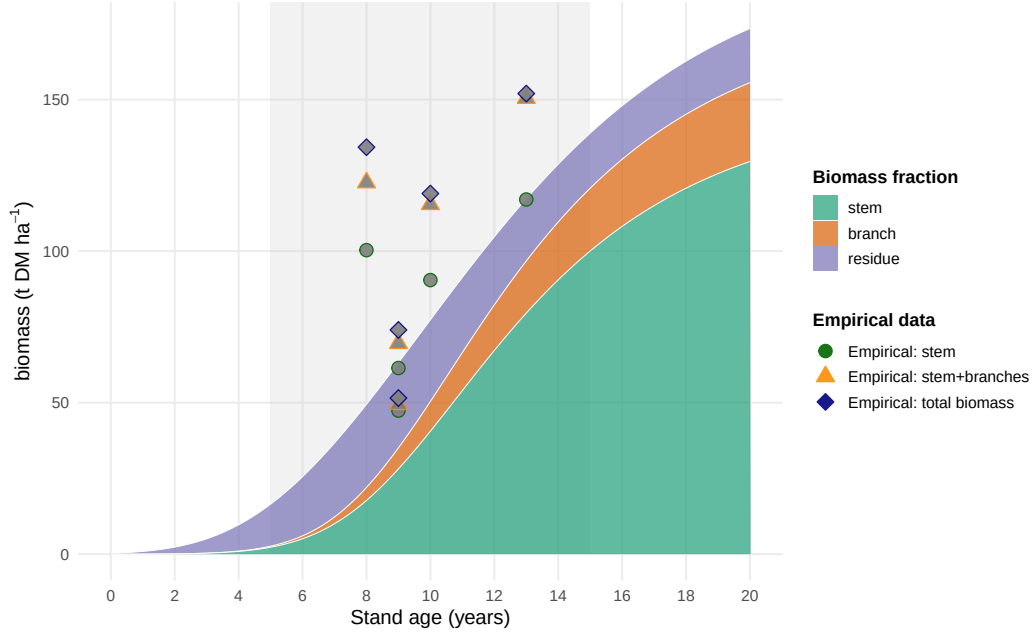


Figure 3: Stand-level aboveground biomass components for poplar under AFS management with $N = 250$ trees ha^{-1} . Empirical AGB observations compiled from Jha (2018), Civitarese et al. (2019), and Niemczyk and Thomas (2021). Grey band: typical medium-rotation harvest window (5–15 years).

the moisture content at felling $\omega = 0.55$ is taken from Civitarese et al. (2019). Figure 3 shows the fractions' stacked biomass quantities $\eta_p(t)$ depending on stand age compared with empirical observations from Jha (2018)

2.4. Biomass supply chain design

The design and planning of biomass supply chains has attracted substantial research attention over the past two decades, driven by the need to ensure cost-efficient and sustainable feedstock provision for bioenergy and biorefinery applications. Several systematic reviews provide comprehensive overviews of this literature. Cambero and Sowlati (2014) survey economic, social, and environmental assessment and optimisation studies for forest biomass supply chains, identifying MILP as the dominant modelling approach and noting a growing trend towards multi-objective formulations integrating life cycle assessment.

Yue et al. (2014) review biomass-to-bioenergy and biofuel supply chain models with a focus on multi-scale modelling, uncertainty treatment, and sustainability metrics, and conclude that the coupling of upstream feedstock dynamics with downstream network design remains an underexplored area. Malladi and Sowlati (2018) survey logistics-specific features of biomass supply chain models—seasonal availability, moisture content degradation, preprocessing steps—and highlight that most models treat biomass as a homogeneous commodity, neglecting quality differentiation. Zahraee et al. (2020) provide a recent meta-review of 140+ studies, confirming that cost minimisation is still the dominant objective, and that multi-product cascading formulations are rare.

The structure of biomass supply chain models typically encompasses three decision layers: (i) strategic network design (facility location, capacity sizing, technology selection), (ii) tactical planning (harvest scheduling, seasonal flows, inventory), and (iii) operational routing. The following paragraphs discuss key individual contributions and their specific methodological advances.

Ekşioğlu et al. (2009) develop one of the earliest large-scale MILP models for biomass-to-biorefinery supply chain design, integrating facility location, capacity sizing, and multimodal transport for lignocellulosic feedstocks. Their model minimises total system cost and introduces binary facility-opening variables that have since become standard in the field. A key limitation, however, is that biomass supply is treated as an exogenous input, so feedstock availability is not endogenised as a function of land-use or growth decisions. Mansoornejad et al. (2013) extend the strategic design perspective to forest biorefineries by integrating product portfolio design with supply chain configuration in a single MILP. Their model explicitly accounts for product conversion efficiencies along different processing pathways, enabling joint optimisation of process technology selection and logistics.

De Meyer et al. (2014) introduce the OPTIMASS framework, a generic MILP for biomass-based supply chain optimisation that explicitly models product quality cascades: higher-quality biomass classes can substitute for lower-quality demands downstream, whereas the reverse is infeasible. This cascade hierarchy, formalised through ordered product sets, is particularly relevant for woody biomass where stem, branch, and residue fractions command different market prices. The companion t-OPTIMASS

extension (De Meyer et al., 2016) incorporates temporal dynamics by explicitly representing biomass growth and regeneration over multiple periods, enabling simultaneous optimisation of harvest timing and network flows. However, in both OPTIMASS formulations the growth dynamics are represented as exogenous time-series rather than as age-structured stand models, so rotation-length decisions and age-dependent biomass quality are not endogenised. Arabi et al. (2019) demonstrate multi-period MILP extensions for bio-based supply chains with production dynamics, but focus on microalgae rather than woody biomass and do not address product quality cascades.

Sharma et al. (2013) identify supply-side uncertainty (yield variability, seasonal availability, moisture content) as the most critical source of risk in biomass supply chains. In response, a strand of literature has developed stochastic and robust optimisation extensions. Zahraee et al. (2020) document that two-stage stochastic MILP and robust optimisation approaches now account for roughly 30% of the modelling literature, with biomass yield uncertainty and price volatility as the dominant stochastic parameters.

Biomass supply chain research focusing specifically on agroforestry or SRC contexts is comparatively sparse. Espinoza et al. (2017) note that most supply chain models assume continuous feedstock availability and do not represent the periodic harvest cycles characteristic of coppice systems. Lamers et al. (2011) analyse feedstock design principles for lignocellulosic supply chains and argue that the allocation of different biomass fractions to valorisation pathways—product cascading—is a key lever for economic viability, yet is rarely optimised jointly with network design decisions. The model developed in Section 3 of this paper directly addresses this gap by endogenising stand age dynamics through binary age-lag variables and coupling component-specific yield coefficients with a product quality cascade.

Table 1 summarises the key modelling features of the reviewed contributions.

3. MILP Model for Agroforestry Supply Chain Design

Building on the biomass growth models introduced in Section 2.3 and the supply chain design literature reviewed in Section 2.4, this section describes the AFS supply chain design problem (AFS-SCD), a mixed-integer linear programming (MILP) model for the integrated design and operation of an agroforestry biomass supply chain with

Table 1: Key modelling features of selected biomass supply chain design studies.

Study	Objective	Multi-product	Quality cascade	Age dynamics	Uncertainty	Level
Eksioglu et al. (2009)	Cost min.	—	—	—	—	Strategic
Sharma & Taaffe (2013)	Cost min.	Partial	—	—	—	Strategic/Tactical
Mansoornejad et al. (2013)	Profit max.	✓	—	—	—	Strategic
Cambero & Sowlati (2014)	Cost/GHG/jobs	—	—	—	Stochastic	Strategic
Yue et al. (2014)	Cost/GHG	✓	—	—	—	Multi-scale
De Meyer et al. (2015)	Cost min.	✓	✓	Exogenous	—	Strategic/Tactical
De Meyer et al. (2016)	Cost min.	✓	✓	Exogenous TS	—	Multi-period
Lamers et al. (2015)	Cost/quality	✓	Partial	—	Scenario	Strategic
Espinoza et al. (2017)	Sustainability	✓	—	—	Scenario	Strategic
Malladi & Sowlati (2018)	Cost min.	Partial	—	—	—	Tactical/Oper.
Arabi et al. (2019)	Profit max.	✓	—	Exogenous	—	Multi-period
Zahraee et al. (2020)	Multi-obj.	✓	—	—	Stochastic	Strategic/Tactical
This paper	NPV max.	✓	✓	Endogenous	Scenario	Strategic/Multi-period

multiple AFS biomass types (i.e., stem, branch, and residue fractions) used as feedstocks for potential consumers in chemical, pulp and paper as well as energy industry.

3.1. Sets, Indices, and Time Structure

The model considers a set of agroforestry sites $\mathcal{I}$, storage/pre-treatment facilities $\mathcal{J}$, consumer sites $\mathcal{K}$, and biomass types $\mathcal{P} = \{stem, branches, residues\}$, whose growth models are outlined above. Time is discretised into annual periods $t \in \mathcal{T} = \{1, \dots, T_{\max}\}$. To reduce complexity, reflect legal definitions of SRAFS and LRAFS as well as discard implausible options (e.g. extremely long rotation cycles), $A^{\min}$ and $A^{\max}$ define minimum and maximum stand ages. Thus, harvest are possible in periods $\mathcal{T}^{\text{harv}} = \{A^{\min} + 1, \dots, T_{\max}\}$. To consistently model establishment and consecutive harvest decisions, the following arc sets are defined: $\mathcal{S}^{\text{est}}$ defines potential establishments of AFS. If an AFS is established in period $t \in \mathcal{T}$, arc $(0, t)$ is selected. Harvest set $\mathcal{S}^{\text{harv}}$ summarizes all potential consecutive harvest periods within the age interval $[A^{\min}, A^{\max}]$. I.e., if an AFS is established

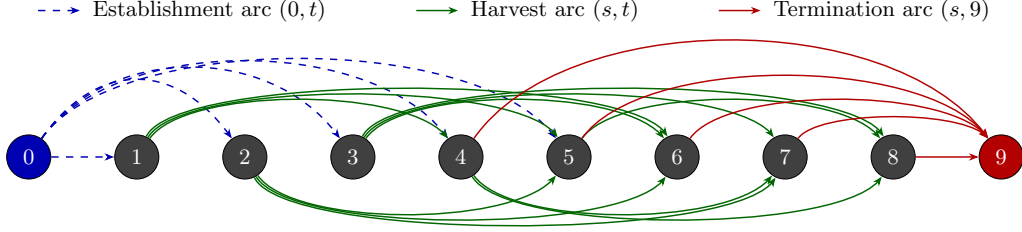


Figure 4: Network structure of set $\mathcal{S}$ for $T = 8$, $A_{\min} = 3$, $A_{\max} = 5$. Dashed blue arcs: establishment arcs $(0, t)$, $t \in \{1, \dots, T - 1\}$. Solid green arcs above: harvest arcs (s, t) with odd s ; below: harvest arcs with even s . Solid red arcs: termination arcs $(s, T + 1)$, $s \in \{A_{\min} + 1, \dots, T\}$.

or harvested in period s and next harvest is in period t , arc (s, t) is selected. Finally, if there is no subsequent harvest after period t , the AFS is terminated by selecting arc $(t, T^{\max} + 1)$. Termination arcs are summarized in set $\mathcal{S}^{term}$. All arc sets are formally defined as follows:

$$\begin{aligned}\mathcal{S}^{est} &= \{(0, t) \mid t \in \{1, \dots, T^{\max} - A^{\min}\}\}, \\ \mathcal{S}^{harv} &= \{(s, t) \mid s, t \in \mathcal{T} \wedge t - s \in \{A^{\min}, \dots, A^{\max}\}\}, \\ \mathcal{S}^{term} &= \{(s, T^{\max} + 1) \mid s \in \{A^{\min} + 1, \dots, T_{\max}\}\}, \\ \mathcal{S} &= \mathcal{S}^{est} \cup \mathcal{S}^{harv} \cup \mathcal{S}^{term}.\end{aligned}$$

Figure 4 illustrates this network structure of all arc sets for $T = 8$, $A^{\min} = 3$, $A^{\max} = 5$.

3.2. Decision Variables

Temporal flow Variables z_{ist} indicate on which quantity of the area of site i an AFS is operated between $(s, t) \in \mathcal{S}$. Depending on the arc type, this implies establishment, harvest, or termination. Selected arcs build a network flow over the complete planning horizon from $t = 0$ to $T^{\max} + 1$. Figure 5 illustrates a simple network flow with establishment of an AFS in period 1 and harvests in periods 5 and 8.

Spatial flow variables X_{ijpt} and $X_{jkpp't}$ represent transport of products from AFS sites to storage/pre-treatment facilities and from there to consumers, respectively, while S_{jpt} captures end-of-period inventory at facility j . Table 2 summarises all sets, decision variables, and parameters used in the MILP formulation, while η_{pa} refers to (4).

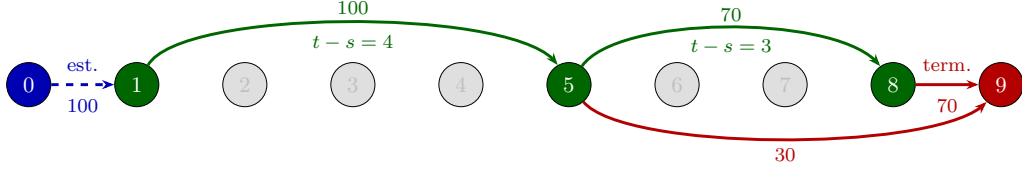


Figure 5: Example of a feasible path through the network for $T = 8, A_{\min} = 3, A_{\max} = 5$: establishment in period 1, harvests in periods 5 and 8, termination at $T + 1$.

3.3. Objective Function

As discussed in Section 2.1, the economic attractiveness of AFS supply chains is modelled from a global perspective taking into account all processes, activities, and participants involved in finally serving customer markets with AFS-based feedstocks. Therefore, revenues for the different biomass fractions and are weighed against costs occurring at all value-adding stages of the supply chains. Revenues created by selling AFS-based feedstocks to different industrial consumers are counterweighted by establishment, opportunity, harvest, transport, and storage costs of landowners, agricultural firms as well as biomass logistics companies. The model therefore maximizes total profit of all participants in the AFS supply chain over the planning horizon as follows:

$$\begin{aligned}
 \max Z = & \sum_{k \in \mathcal{K}} \sum_{p \in \mathcal{P}} R_{kp} \cdot \sum_{j \in \mathcal{J}} \sum_{(p', p) \in \mathcal{Q}} \sum_{t \in \mathcal{T}} X_{jkp'pt} \\
 & - \sum_{i \in \mathcal{I}} c_i^{est} \cdot \sum_{(0, t) \in \mathcal{S}^{est}} z_{i0t} \\
 & - \sum_{i \in \mathcal{I}} (c_i^{main} + c_i^{opp}) \cdot (t - s) \cdot \sum_{(s, t) \in \mathcal{S}^{harv}} z_{ist} \\
 & - \sum_{i \in \mathcal{I}} c_i^{harv} \cdot \sum_{(s, t) \in \mathcal{S}^{harv}} z_{ist} \\
 & - \sum_{i \in \mathcal{I}} \sum_{j \in \mathcal{J}} \sum_{p \in \mathcal{P}} \sum_{t \in \mathcal{T}} c_p^{tr-raw} \cdot d_{ij} \cdot X_{ijpt} \\
 & - \sum_{j \in \mathcal{J}} \sum_{k \in \mathcal{K}} \sum_{(p, p') \in \mathcal{Q}} \sum_{t \in \mathcal{T}} c_{p'}^{tr-pre} \cdot d_{jk} \cdot X_{jkpp't} \\
 & - \sum_{j \in \mathcal{J}} \sum_{p \in \mathcal{P}} \sum_{t \in \mathcal{T}} c_j^{stor} \cdot S_{jpt}
 \end{aligned} \tag{5}$$

Here, R_{kp} are product- and consumer-specific revenues for each ton of biomass fraction (€/t). The opportunity cost term penalises the land use when AFS are established.

Table 2: Notation for the MILP model.

Sets & Indices	
$i \in \mathcal{I}$	Agroforestry sites
$j \in \mathcal{J}$	Storage / pre-treatment facilities
$k \in \mathcal{K}$	Consumer sites
$p \in \mathcal{P}$	Product types (stem, branches, residues)
$(p, p') \in \mathcal{Q}$	Admissible product-substitution pairs (quality cascade)
$\mathcal{T} = \{1, \dots, T\}$	Planning periods (years)
$\mathcal{T}^{\text{harv}} = \{A^{\min} + 1, \dots, T\}$	Feasible harvest periods
$\mathcal{S}^{\text{est}} = \{(0, t) \mid t \in \{1, \dots, T - A^{\min}\}\}$	Establishment arcs: dummy source to period t
$\mathcal{S}^{\text{harv}} = \{(s, t) \mid s, t \in \mathcal{T}, t - s \in \{A^{\min}, \dots, A^{\max}\}\}$	Harvest cycles within age-lag bounds
$\mathcal{S}^{\text{term}} = \{(s, T + 1) \mid s \in \{A^{\min} + 1, \dots, T\}\}$	Termination arcs to dummy sink $T + 1$
$\mathcal{S} = \mathcal{S}^{\text{est}} \cup \mathcal{S}^{\text{harv}} \cup \mathcal{S}^{\text{term}}$	Age-lag arc set (establishment $\cup$ harvest $\cup$ termination)
Decision Variables	
$z_{ist} \geq 0$	Area of site i with AFS operation between s and t
$X_{ijpt} \geq 0$	Transport flow (t) of p from site i to facility j in t
$X_{jkpp't} \geq 0$	Flow of p from facility j to consumer k satisfying demand p' in t
$S_{jpt} \geq 0$	Inventory (t) of product p at facility j end of period t
Parameters	
T	Length of planning horizon (years)
$A^{\min}, A^{\max}$	Minimum / maximum stand age at harvest (years)
AREA_i	Area of site i (ha)
η_{pa}	Yield coefficient (t/ha) of fraction p at stand age a , derived from (4)
R_{kp}	Revenue (€/t) for product p at consumer k
C_i^{est}	Establishment cost (€/ha) at site i
C_i^{opp}	Annual opportunity cost (€/ha) at site i
C_i^{harv}	Harvest & refitting cost (€/ha) at site i
$c^{\text{tr-raw}}$	Transport cost rate (€/t km) for raw biomass
$c^{\text{tr-pre}}$	Transport cost rate (€/t km) for pre-treated biomass
c_j^{stor}	Storage holding cost (€/t) at facility j
d_{ij}, d_{jk}	Distance (km) between nodes
$\text{CAP}_j^{\text{stor}} / \text{CAP}_j^{\text{proc}}$	Storage and processing capacity (t) at facility j
$D_{kpt}^{\max}$	Maximum demand (t) of fraction p at consumer k in period t

348 Thereby, c_i^{opp} can be negative or positive depending on the site-specific conditions. It
349 comprises effect of lost field crop revenues due to AFS as well as higher yields of remaining

field crop areas e.g. due to wind and water erosion protection by AFS. While the assessment of lost field crop revenues is usually well predictable, the economic evaluation of positive ecological effects of AFS (e.g. on neighboring crop yields) is more difficult. Thus, c_i^{opp} can also contain regulatory subsidies to support establishing and maintaining AFS. Transport cost terms mirror the two-stage flow structure of the AFS supply chain with differentiated rates for raw and pre-treated biomass depending on the biomass type. Cost rates differ because as e.g. the density and water content of the shipped biomass types differ or because specialized equipment (e.g. for log transports) is used [REFERENCES]. Moreover, site-specific establishment cost for planting the AFS trees is considered, next to yearly maintenance and harvesting costs. All these costs scale with the size of a site ($AREA_i$) measured in hectar. Maintenance cost comprise e.g. costs for trimming, re-planting, and pruning unnecessary shoots. Storage costs are considered if biomass types are stored for more than a year. Storage cost rate c_j^{stor} measures the costs for storing a ton of biomass for one year and comprises direct storage cost (e.g. for storage site maintenance) as well as opportunity cost due to potential material losses and capital commitment. Biomass types harvested and consumed in the same year incur no storage cost, as usually AFS are harvested at the beginning of a year. Harvested biomass has to dry during spring and summer such that the biomass can be used in autumn or winter the same year. As under-year drying is necessary for some industrial applications and beneficial for transport economics, drying storage in the year of harvest is not associated with additional cost in this study.

3.4. Constraints

Constraints (6) limits the area dedicated to AFS operations to the maximum size of site i . Constraints (7) enforce flow conservation in each period. If AFS systems of size z_{ist} are established or harvested in period $t \in \mathcal{T}$, there needs to be a succeeding activity (harvest or termination) of this area. This way, (7) guarantee valid (possibly empty) AFS activity over the complete time horizon for each site. Constraints (8) links the harvest-cycle variables z_{ist} to physical yield quantities $\eta_{p(t-s)}^{fw}$ for age $t - s$, which are pre-computed using (1)–(4) scaled by moisture content ω . Thus, we assume constant yields per hectar for each site i . Note that this simplification can be dropped and site-specific yields $\eta_{ip(t-s)}^{fw}$ could be introduced without increasing numerical complexity of the AFS-

SCD . Constraints (9) link harvest quantities to transport flows and ensure inventory balance at each facility. Thereby, harvest quantities can be smaller than available AFS biomass allowing partial harvesting. Constraints (10)–(11) enforce storage and processing capacity limits at each facility j . Constraints (12) implement product usage at multiple consumption sites k . Thereby, $(p', p) \in \mathcal{Q}$ defines a product mapping allowing high-quality product p' (e.g. stems) satisfying demand of a lower-quality products p up to maximum demand $D_{kpt}^{\max}$.

$$\sum_{(0,t) \in \mathcal{S}^{est}} z_{i0t} \leq AREA_i \quad \forall i \in \mathcal{I} \quad (6)$$

$$\sum_{(s,t) \in \mathcal{S}} z_{ist} = \sum_{(t,u) \in \mathcal{S}} z_{itu} \quad \forall i \in \mathcal{I}, t \in \mathcal{T} \quad (7)$$

$$\sum_{j \in \mathcal{J}} X_{ijpt} \leq \sum_{(s,t) \in \mathcal{S}^{harv}} \eta_{p(t-s)}^{fw} \cdot z_{ist} \quad \forall i \in \mathcal{I}, p \in \mathcal{P}, t \in \mathcal{T}^{harv} \quad (8)$$

$$S_{jpt} = S_{jp,t-1} + \sum_{i \in \mathcal{I}} X_{ijpt} - \sum_{k \in \mathcal{K}, (p,p') \in \mathcal{Q}} X_{jkpp't} \quad \forall j \in \mathcal{J}, p \in \mathcal{P}, t \in \mathcal{T}^{harv} \quad (9)$$

$$\sum_{p \in \mathcal{P}} S_{jpt} \leq CAP_j^{stor} \quad \forall j \in \mathcal{J}, t \in \mathcal{T} \quad (10)$$

$$\sum_{i \in \mathcal{I}, p \in \mathcal{P}} X_{ijpt} \leq CAP_j^{proc} \quad \forall j \in \mathcal{J}, t \in \mathcal{T}^{harv} \quad (11)$$

$$\sum_{j \in \mathcal{J}, (p',p) \in \mathcal{Q}} X_{jkp'pt} \leq D_{kpt}^{\max} \quad \forall k \in \mathcal{K}, p \in \mathcal{P}, t \in \mathcal{T}^{harv} \quad (12)$$

$$z_{ist}, X_{ijpt}, X_{jkp'pt}, S_{jpt} \geq 0 \quad \forall i, j, k, p, p', t \quad (13)$$

In summary, AFS-SCD constitutes are large-scaled LP which allows to decide on which fraction of a site, AFS are established and how the AFS operations are scheduled in a multi-year planning horizon. The AFS-SCD takes a supply chain perspective and maximizes total profit of all actors involved in these processes. Thus, it does not explain how the profits are split e.g. between farmers, AFS service providers and logistics service providers. The continuous modelling of AFS areas in z_{ist} leads to a comparatively easy to solve problem. On the other side, it is unclear which part of a site's total area is chosen for AFS operations if $z_{ist} < AREA_i$, which holds for establishment as well as harvesting and termination.

397 4. Case Study

398 The AFS-SCD is intended to support the design of regional agroforestry biomass sup-
399 ply chains fostering regional bioeconomy clusters thereby connecting multiple industrial
400 sectors. The proposed AFS-SCD model is instantiated on a real-world biomass supply
401 chain case study located in the tri-state region of Saxony-Anhalt, northern Thuringia,
402 and northern Saxony, Germany. This region is characterized by a high density of agri-
403 cultural land usage as well as cluster of paper and wood as well as chemical industry.
404 The regional chemical industry relies on fossil resources so far, but is under substantial
405 transformative pressure which led to investments in sustainable production processes
406 requiring sustainable regional bio-based feedstocks. Figure 6 shows agricultural produc-
407 tion sites along with the set of potential wood-processing hubs, and the regional users of
408 AFS biomasses from chemical, paper and wood processing industry as well as the energy
409 sector.

410 The supply chain comprises 2239 agricultural production parcels which are grouped
411 into $|\mathcal{I}| = 145$ potential AFS production sites. Clustering is based on geographical dis-
412 tances between parcel centroid using a k-means approach in order to reduce problem
413 size. For distance calculations, for each site a centroid is calculated. Pre-treatment and
414 intermediate storage facilities $\mathcal{J}$ are located on existing wood processing facilities as well
415 as potential new log or wood chip yards. All harvested biomass quantities at AFS sites
416 are transported to logistics hubs first. At the hubs sorting and transshipment processes
417 take place before biomass types are forwarded to the consumer facilities. Storing and
418 transshipment capacities at the hubs are assumed to be constant in time and universally
419 set to 4.0909×10^4 kt per year. Finally, $|\mathcal{K}| = 28$ consumer sites are included. Each con-
420 sumer site belongs to an existing facility from chemical, paper/wood processing industry,
421 or the energy sector. Distances between the facilities d_{ij} and d_{jk} are calculated as road
422 distances between the facilities' coordinates. For each consumer facility, an yearly upper
423 demand capacity for each of the three distinguished biomass product types are given
424 along with consumer-specific prices. The following biomass types are distinguished:

- 425 • **P1: stem wood – chemical/paper grade:** long-fibre wood chips from stems
426 suitable for conversion in a local biorefinery and paper factory. In total, a yearly
427 demand of 620 kt and an average prices of 45 € is assumed.

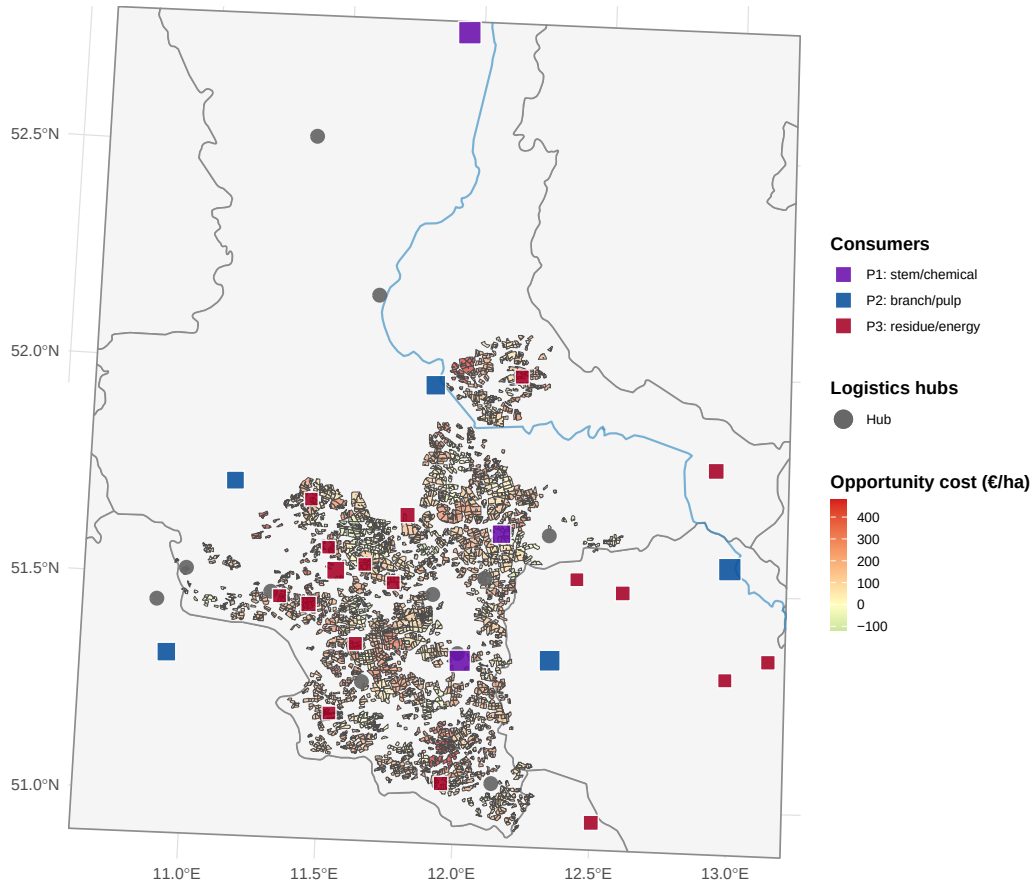


Figure 6: AFS supply chain network in Saxony-Anhalt. Grey lines: German federal state boundaries. Blue lines: major rivers. Coloured polygons: potential AFS parcels grouped by site (colours encode cluster membership; no legend). Orange symbols: pre-treatment facilities (circle = sawmill, triangle = logistics hub, size proportional to processing capacity). Coloured squares: consumer sites (violet: P1 chemical/pulp, blue: P2 pulp/paper, red: P3 energy/biogas, size proportional to total demand).

- 428 • **P2: branch wood – sawmill/pulp grade:** medium/short-fibre wood chips
429 from larger branches suitable for mechanical processing in sawmills and lower-grade
430 pulping. Demand is distributed across 6 smaller facilities and accumulates to 410
431 kt/year. Gate prices range from 28-38 €/t.
- 432 • **P3: residue – energy grade:** residual wood fractions (chips, bark, off-cuts) used
433 for heat and electricity generation or lower-grade pellet production. Demand is
434 accumulates among 24 consumers to 132 kt/year. Gate prices range from 18-24
435 €/t.

436 Table 5 in the appendix summarizes the consumer-specific demand data. It is assumed
437 that the biomass product types show a quality cascade which means that the demand of
438 pulp grade (P2) can also be satisfied by chemical/paper grade biomass (P1). Likewise,
439 energy grade demand (P3) can be satisfied by all biomass types. Note that constant
440 demands over the planning horizon are assumed, although an increasing biomass demand
441 can be expected due to the green transition in those industries. From this perspective, a
442 rather conservative market demand scenario is assumed.

443 Growth model introduced in subsection 2.3 is used to model biomass quantities de-
444 pending on age and the area dedicated to AFS. The growth model is parameterized for
445 poplar clones at central European conditions as introduced above with a planting density
446 of $N = 400$ trees per hectar. If AFS are established, they have to be harvested within
447 $A^{min} = 3$ to $A^{max} = 20$ years allowing a wide range of potential harvesting strategies
448 between short-rotation and long-rotation AFS. As the total planning horizon is $|\mathcal{T}| = 40$
449 years multiple harvesting cycles can be established. The sites do not differ with respect
450 to growth conditions as the climatic and soil quality variation within the comparatively
451 small region is rather limited. However, the sites differ with respect to size (given in
452 hectar) and opportunity cost rates (given in € per hectar and year). Site-specific oppor-
453 tunity cost rates c_i^{opp} is normally distributed with mean 521 and standard deviation 391.
454 For most sites, substantial opportunity costs are assumed reflecting the loss in field-crop
455 revenue, risk premiums of landowners as well as higher costs for field crop maintenance.
456 The 145 sites with negative opportunity cost rates reflect sites where AFS lead to an
457 increase in field crop productivity on the remaining field area. The variation in opportu-
458 nity cost rates to study the effect of opportunity cost on site selection for AFS and gives

valuable insights particularly for communication with farmers and politics. Subsidies for establishing AFS are one of the core instruments to foster AFS. The height and effectiveness of those subsidies, however, is controversially disputed. Remaining cost parameters are universally set for all sites and comprise maintenance cost rates c_i^{main} (€ per year and hectare), establishment cost c_i^{est} (€ per hectare) and harvesting cost c_i^{harv} (€ per hectare). Likewise, transport cost rates for harvested biomasses c_p^{tr-raw} and pre-treated/sorted biomasses $c_{p'}^{tr-raw}$ are set universally. All scalar parameters are shown in Table 3.

Table 3: Parameter values for the AFS-SCD case study. Transport costs refer to road distance

Parameter	Description	Base-case value	Unit
Index sets			
$ \mathcal{T} $	Planning periods	40	yr
$ \mathcal{I} $	Sites	145	–
$ \mathcal{J} $	Pre-treatment / storage nodes	11	–
$ \mathcal{K} $	Consumers	28	–
$ \mathcal{P} $	Product grades	3	–
Biomass dynamics			
$A^{\min}$	Minimum harvest age (years)	3	yr
$A^{\max}$	Maximum stand age (years)	20	yr
Cost parameters			
c^{tr-raw}	Raw transport cost	0.08	€ t ⁻¹ km ⁻¹
c^{tr-pre}	Pre-processed transport cost	0.06	€ t ⁻¹ km ⁻¹
c_i^{est}	Establishment cost rate	2500	€ ha ⁻¹ yr ⁻¹
c_i^{harv}	harvest cost rate	150	€ ha ⁻¹ yr ⁻¹
c_i^{main}	Maintenance cost rate	10	€ ha ⁻¹ yr ⁻¹
c_i^{opp}	aver. opportunity cost	521	€ ha ⁻¹ yr ⁻¹
Revenue parameters			
R_{P1}	Mean gate price P1	45	€ t ⁻¹
R_{P2}	Mean gate price P2	33	€ t ⁻¹
R_{P3}	Mean gate price P3	21	€ t ⁻¹

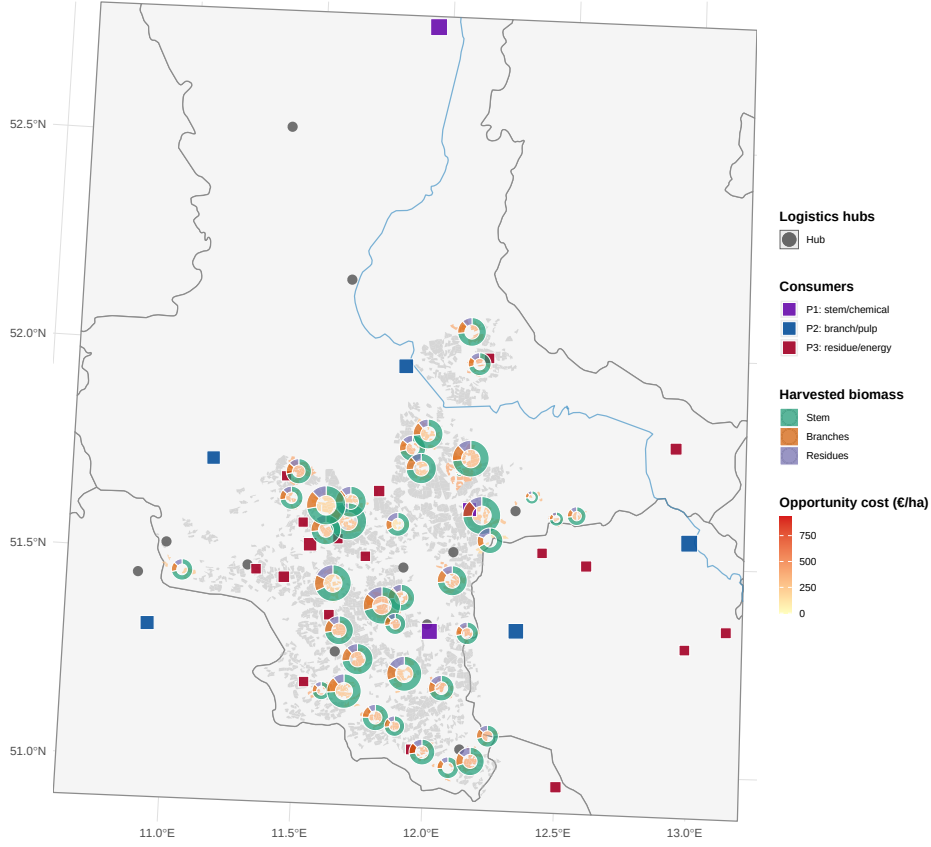


Figure 7: Site selection and biomass production in optimal supply chain configuration. Green polygons: active sites with biomass harvest; grey polygons: non-selected sites. Pie charts at active sites show the product mix of harvested biomass, scaled by total harvested volume.

4.1. Optimization results

The AFS-SCD model is solved for the instance data described above with Gurobi. The model selects a subset of sites for AFS establishment and identifies the optimal harvest schedules, intermediate facility utilisation, and product allocation across the planning horizon.

Figure 7 shows the spatial configuration of the profit-optimal supply chain.

In total, on 38 sites AFS are established which corresponds to about 26 % of all sites and 24 % of the total farm area. Over the planning horizon, 2.6368×10^4 kt of AFS biomasses are produced on these sites which suffices to satisfy 57 % of total demand. Figure 8 shows the harvested biomasses over time along with total demand of

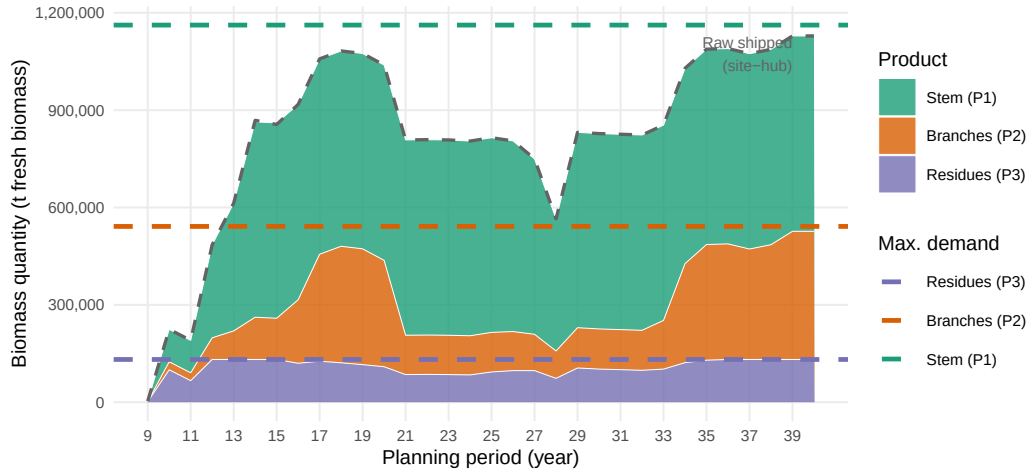


Figure 8: Total biomass delivered to consumers by product type over the planning horizon. Stacked areas show period-level volumes of stem (P1), branches (P2), and residues (P3) flowing from storage hubs to all consumers. Dashed line: total harvested raw biomass shipped from sites to hubs.

all consumer sites.

During the first 10 years no AFS biomasses are harvested as trees of established AFS in first years grow up to a certain size before first harvest. Thus, about 25% of consumers' total biomass demand cannot be satisfied due to this grow-up period. Afterwards, a cyclical pattern appears with a harvesting peak around year 20 and a breakdown in year 28 before plateau is reached from year 35-40. Thereby, the harvesting quantities of stems and branches show a similar pattern while quantities of harvested energy grade biomass are rather stable and close to the demand limit over the planning horizon. Thus, rather long-rotation AFS are established as Figure 9 confirms showing the histogram of harvesting ages across all sites.

The histogram illustrates that majority of harvest events occur at stand ages between 14-18 years, indicating that long-rotation AFS seem to be more profitable in this setting. Short-rotation cycles (<10 years) occur only in marginal frequencies. This optimal rotation length might be caused either due to the limited demand of energy-grade biomass or the price gaps between biomass types or a combination of effects. The quantity of energy-grade biomass created in long-rotation AFS suffices to cover total energy-grade biomass demand almost completely. Rotation length is focused on production of chemical-grade

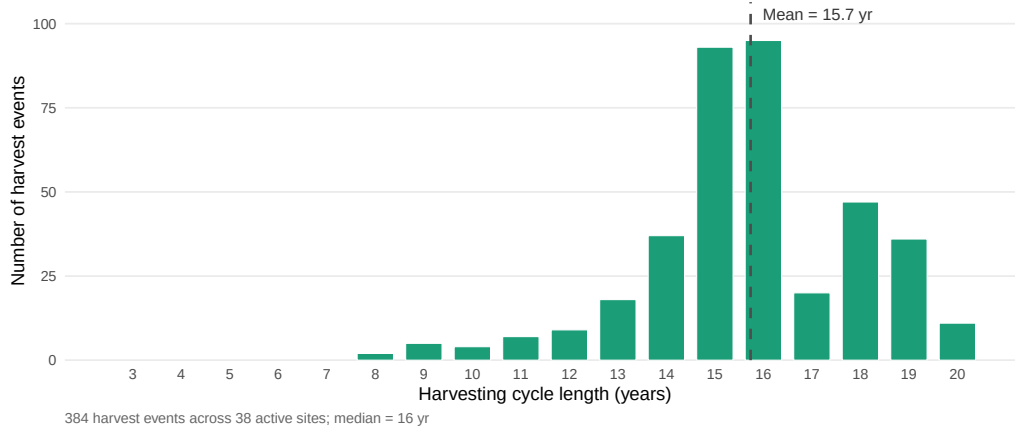


Figure 9: Distribution of harvesting cycle lengths at active agroforestry sites. Each bar represents the number of individual harvest events (over all periods and active sites) whose cycle length (stand age at harvest, i.e. $t - s$) falls in that year bin.

493 biomass where highest prices are assumed. Focusing on chemical industry supply requires
 494 a production of trees with a comparatively large stems. This strategy maximizes revenue,
 495 but in the same time increases opportunity costs as wood harvests appear in larger time
 496 intervals. Figure 10 presents the aggregate cost-profit waterfall over all periods and
 497 actors.

498 It appears that revenue is dominated by selling chemical-grade biomass and domi-
 499 nating cost component are opportunity cost although active sites are those sites with
 500 a comparatively small opportunity cost rate. Next to opportunity cost, establishment
 501 and transport costs are the most relevant cost drivers. Analyzing total cashflow over
 502 time shows that break-even is reached after about 20 years, i.e. after the first large-scale
 503 harvest occurred. Afterwards, cashflow is continuously increasing indicating that estab-
 504 lishing AFS is a strategic change in agricultural practice and pays only if this strategy
 505 is implemented and maintained over a comparatively long time spans. This underpins
 506 the importance of strategic buying contracts with industrial partners in order to reduce
 507 revenue risks of farmers when producing AFS biomass.

508 Although profitability of the total AFS supply chain is given for the case study data,
 509 profitability of AFS operations on site-level are more heterogeneous. Figure 11 shows
 510 revenues and costs split by component and normalized to AFS size and active years of

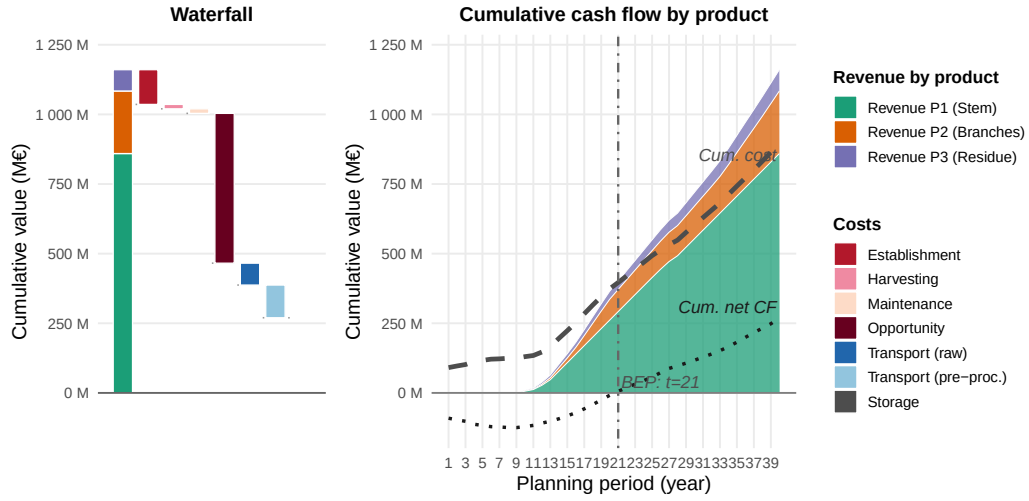


Figure 10: Total cost–profit breakdown across all actors and years. The left panel (waterfall) decomposes the objective value into revenue and individual cost categories; the right panel shows the period-level trajectory of revenues and stacked costs.

511 AFS.

512 It appears that net profitability of AFS ranges between 282 and 6 € per hectare
513 and active year. Note that this value constitutes additional profit after subtracting
514 operational cost as well as imputed opportunity cost of lost profit from field crop. I.e.,
515 for calculating the realized net margin, opportunity can be added such that realized
516 yearly net profit ranges between 473 and 210 € per hectare. Still, relative net profit
517 is comparatively low compared to ordinary agricultural practices with profitable field
518 crops (like winter wheat). The revenue splits of active sites indicates that at some sites
519 medium-grade and low-grade biomass types contribute substantially or even solely to
520 total revenue (see e.g. site 12, 135, 145, and 112). This indicates that site-specific AFS
521 strategies might differ from the overall supply chain strategy. In other words, there seem
522 to be strategic niches w.r.t. the consumer segments to be supplied. This might also be
523 reflected in the rotation length and, although not discussed here, the selection of trees
524 and clones for the AFS. Moreover, profitability of AFS sites seems to depend on various
525 factors like local opportunity costs as well as closeness to consumers. Figure 12 shows
526 net margins and opportunity cost rates of sites with active AFS.

527 As expected net margins of AFS depend heavily on a site's opportunity cost rate

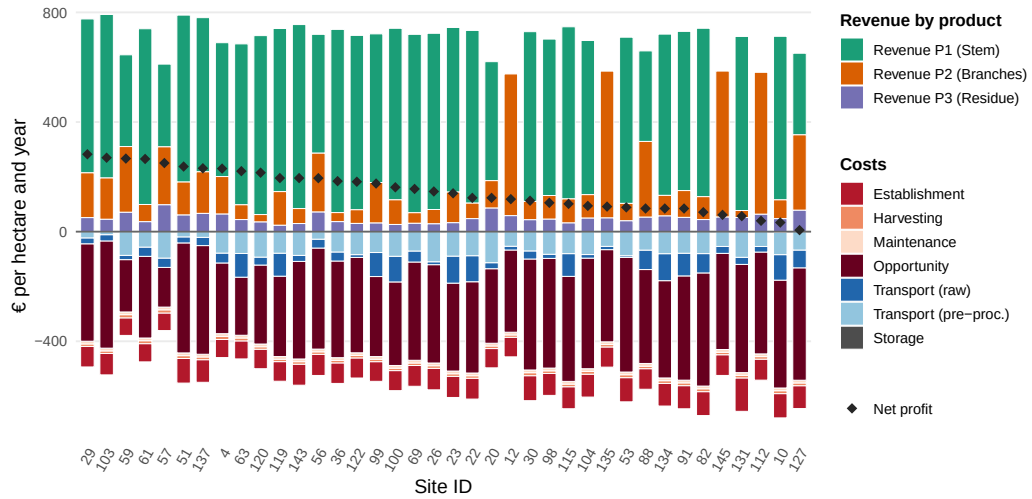


Figure 11: Revenues, costs and net profit for sites with AFS operations split by type and expressed in € per hectare per year of AFS operation.

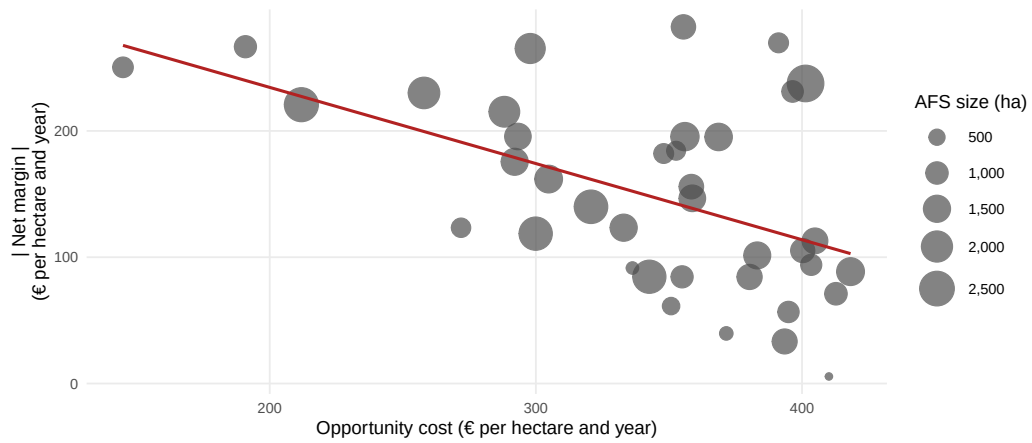


Figure 12: Scatterplot of opportunity cost rate and net margin (both in € per ha and year) for each site with active AFS. Size of AFS reflected in point size.

528 showing a clear negative trend. Moreover, it appears that AFS sizes vary quite heavily
529 between 86 and 2794 hectare. Probably very small sites won't operate efficiently because
530 cost for establishment, harvesting, and maintenance are assumed to scale with size, but
531 are rather fixed below a certain critical threshold.

532 5. Conclusion

533 This paper has developed and applied the AFS-SCD model, a MILP formulation
534 for the integrated design and operation of agroforestry biomass supply chains. The
535 model endogenises stand age dynamics through binary arc-based age-lag variables, cou-
536 ples component-specific Gompertz yield coefficients for stem, branch, and residue frac-
537 tions with multi-echelon logistics decisions, and embeds a product quality cascade that
538 allocates biomass grades to chemical, pulp, and energy markets within a unified opti-
539 misation framework. The formulation is, to the best of our knowledge, the first supply
540 chain model to simultaneously optimise AFS establishment timing, consecutive harvest
541 scheduling under rotation-age constraints, and product-grade allocation across spatially
542 heterogeneous consumer markets.

543 The case study for the Saxony-Anhalt tri-state region demonstrates the practical rel-
544 evance of the approach. The optimal supply chain concentrates AFS establishments in
545 the Elbe and Saale floodplains, where high biomass productivity and proximity to large
546 pulp mills create the most favourable economics. The sensitivity analysis confirms that
547 profitability depends primarily on achievable gate prices for high-quality (P1) biomass
548 grades: scenarios combining reduced revenue with high opportunity costs yield negative
549 net profits, whereas scenarios with full or elevated prices remain viable under cost pres-
550 sure. Opportunity cost is the dominant site-level discriminator, with parcels exceeding
551 approximately $300 \text{ € ha}^{-1} \text{ yr}^{-1}$ rarely selected in the optimal solution. These findings
552 underscore the importance of product cascading as a strategic lever for AFS viability:
553 maximising the share of stem biomass directed to high-value chemical and pulp applica-
554 tions substantially improves profitability compared to energy-only offtake strategies.

555 Methodologically, the AFS-SCD model bridges the gap between stand-level growth
556 modelling and regional supply chain optimisation identified in the literature (Zahraee
557 et al., 2020; Lamers et al., 2011). The age-lag constraint structure generalises to other

perennial biomass crops with periodic coppice cycles, such as willow or black locust, and to long-rotation silvoarable systems in which rotation length is a strategic variable. The model is extensible to stochastic settings by replacing deterministic yield coefficients with scenario-weighted counterparts or by incorporating robust optimisation over yield uncertainty intervals (Sharma et al., 2013).

Several limitations point to directions for future research. First, consumer demand is treated as exogenous and time-invariant; endogenising demand responses to biomass price changes would require a partial-equilibrium extension. Second, the case study uses a single allometric scenario calibrated to Central European poplar; propagating Gompertz parameter uncertainty through to supply chain outcomes would strengthen the robustness analysis. Third, environmental co-benefits of AFS — including carbon sequestration, nitrate leaching reduction, and biodiversity enhancement — are not monetised in the current objective. Integrating life cycle assessment indicators as additional objectives or constraints would support a more comprehensive sustainability evaluation, consistent with the cleaner production framing of this work.

6. Appendix

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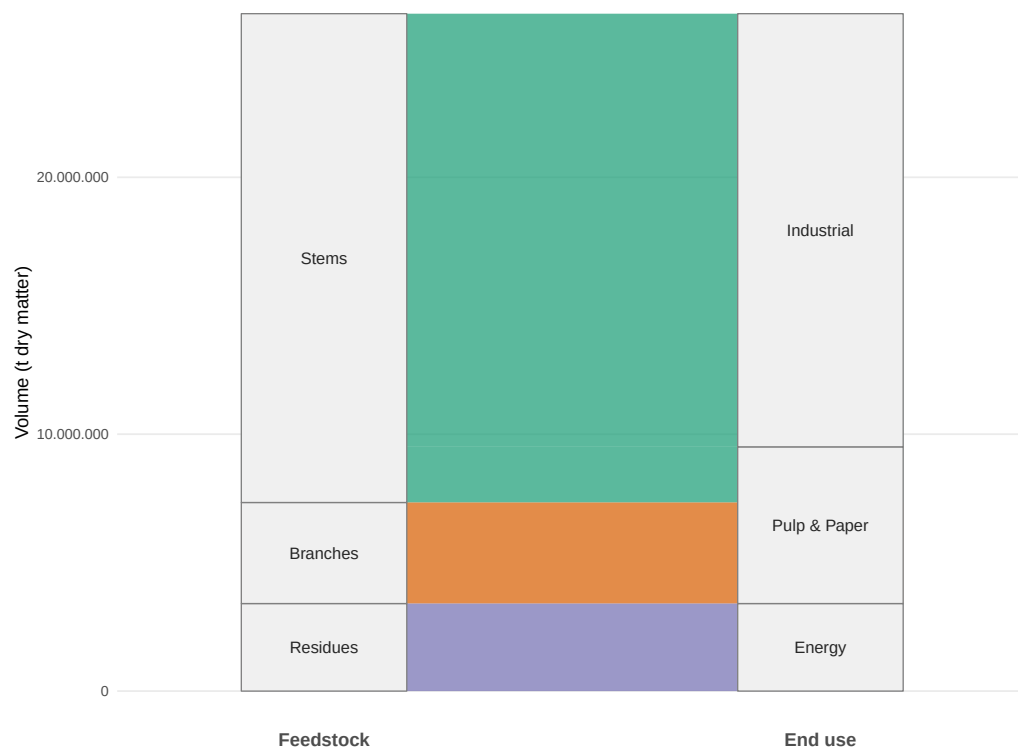


Figure 13: Total production quantity of biomass types and its industrial usage over planning horizon (in tons).

Table 4: Notation for the biomass growth and harvest yield models.

Symbol	Description
$M(t)$	Total above- and belowground dry biomass per ha of AFS (t DM ha ⁻¹) at stand age t
$A(N)$	Stand-level Gompertz asymptote (t DM ha ⁻¹); density- and site-dependent
k	Gompertz growth-rate coefficient
t_0	Inflection age of stand biomass accumulation (years)
N	Stand density (trees ha ⁻¹)
C_{site}	Site-quality constant (kg ha ^{β-1}); conservative: 4 475, good site: 6 471
β	Competition-density exponent; fitted $\hat{\beta} = 0.380$
$f_p(t)$	asymptotic fraction of AGB attributed to compartment $p \in \{s, b, r\}$ at age t
$q_p(t)$	Merchantable fraction of compartment p biomass at age t (logistic)
r_p	Logistic growth rate for merchantability of compartment p
$t_{50,p}$	Age at which 50% of compartment p biomass is merchantable (yr)
$\eta_p(t)$	Merchantable dry biomass of compartment p per ha at harvest age t (t DM ha ⁻¹)
ω	Moisture content at felling (mass fraction); $\omega = 0.55$

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Table 5: Consumer demand quantities and gate prices by product grade. P1: chemical/pulp; P2: sawlog/pulpwood; P3: energy/pellets. Demands in kt wood input per year; prices in €/t wood input at the factory gate.

Consumer	Annual demand			Gate price		
	P1 (kt/yr)	P2 (kt/yr)	P3 (kt/yr)	P1 (€/t)	P2 (€/t)	P3 (€/t)
consumer 1	300.0	–	–	50	–	–
consumer 2	100.0	–	–	60	–	–
consumer 3	–	300	50.0	–	37.5	24
consumer 4	200.0	–	–	47.5	–	–
consumer 5	–	30	5.0	–	35	21
consumer 6	–	15	5.0	–	32.5	21
consumer 7	–	10	3.0	–	30	18
consumer 8	–	5	8.0	–	27.5	18
consumer 9	2.0	50	20.0	40	35	22.8
consumer 10	–	–	1.7	–	–	21
consumer 11	–	–	2.9	–	–	21
consumer 12	–	–	2.5	–	–	21
consumer 13	–	–	2.3	–	–	21
consumer 14	–	–	1.9	–	–	21
consumer 15	–	–	2.7	–	–	21
consumer 16	–	–	1.7	–	–	21
consumer 17	–	–	1.9	–	–	21
consumer 18	–	–	1.6	–	–	21
consumer 19	–	–	1.5	–	–	21
consumer 20	–	–	1.5	–	–	21
consumer 21	–	–	1.8	–	–	21
consumer 22	–	–	3.5	–	–	21
consumer 23	–	–	1.6	–	–	21
consumer 24	–	–	2.1	–	–	21
consumer 25	–	–	4.5	–	–	21
consumer 26	–	–	1.8	–	–	21
consumer 27	–	–	3.2	–	–	21
consumer 28	18.5	–	–	27.5	–	–